# *ALILICHE Larbi*

Embedded Linux Engineer

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CV en Français
https://aliliche.github.io/resume/
github.com/Aliliche



## *Experience*

**Wizzilab** *Avril 2026 - Today*
*Embedded Linux Engineer* *Montrouge, France*

- Maintaining Yocto distribution, various fixes and internal tools enhancement.

**ACKSYS Communications & Systems** *November 2021 - avril*
*Embedded Software Engineer* *Yvelines, France*

◇ Recently worked on:

- Ported the roaming strategy to the new OpenWrt 24 / Linux kernel 6.6: Hostapd, wpa_supplicant, mac80211, ath10k and ath11k drivers.
- In-depth analysis of ath11k (CE, WMI, MAC, DP...) and the mac80211 layer to successfully reduce Wi-Fi roaming time.
- Developed a dual-boot mechanism (U-Boot) with recovery features.
- Designed and implemented a new embedded WebUI for ACKSYS products, migrating from Lua to a fully JavaScript-based architecture.

◇ Previous work:

- Customized Wi-Fi roaming strategy (PLH) : mac80211, ath11k, and wpa_supplicant.
- Project SRCC – Smart Redundancy Carriage Coupling: enhanced reliability through smart redundancy mechanisms
- Developed "surveyd", a monitoring daemon to detect and handle Wi-Fi link failures on ACKSYS routers.
- Maintained WaveOS, the acksys operating system from userspace to kernel: LED management, ath10k/ath11k drivers, mac80211, hostapd/supplicant.

**Tiempo Secure** *February 2021 - jully 2021*
*Intern, Linux driver developer* *Grenoble, France*

1. MTD driver for S25FL Flash memory and Block Driver for MMC :
   - Implemented core driver structure on the TIEMPO Secure Enclave RAM.
   - Used UBI API (UBI File System) for Wear leveling , Garbage collection.
2. Platform driver for TIEMPO IPC (Inter-Processor Communication FPGA Block) :
   - Defined technical specification for the driver.
   - Analyzed and validated the IP in bare-metal prior to the driver developement.
   - Developed the Linux platform driver : Device Tree, interrupt handlers, probe function, char device interface and the mailbox management.
   - Created user-space test tool.
   - Buildroot integration.

**Centre Spatial Universitaire de Grenoble** *November 2020 - February 2021*
*Final Year Project* *Grenoble, France*

- Developed a small FreeRTOS application on Microblaze soft-core: Data acquisition from FPGA memory and sending them via Lightweight TCP/IP stack. The Developement included the hardware implementation of the Microblaze IP blocks and the software application running on top.

## *Education*

**Master II** *Université Grenoble Alpes*
*Micro-électronique, Intégration des systèmes temps réels embarqués* *September 2019 - August 2021*

- Embedded systems.
- RTOS: Interruptions, scheduling, algorithms, optimization(freeRTOS, preemt-rt).
- Modules, device drivers, computer architecture, execution models.

**Master I** *Université Mouloud Mammeri Tizi Ouzou*
*Instrumentation Électronique* *septembre 2014 - Aout 2018*

## *Languages*

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| **English, French, Arabic** | Fluent |
| **Kabyle** | Native |

## *Skills*

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| **Embedded Systems** | Strong knowledge of Embedded Linux, Linux Wireless (OpenWrt). Cross-compilation toolchains and Linux BSP : Yocto, Buildroot, U-Boot. Solid understanding of Linux drivers: Char, Block, MTD, Platform. Driver development for FPGA blocks (RISC-V architecture). Solid understanding of the WI-FI stack. |
| **Manufacturing** | Good understanding of production processes in France, certifications, and interaction with international clients. Strong knowledge of testing, validation, and bug management in production environments. |
| **Languages** | C, Shell, Risc-V asm, Lua, HTML/CSS, JavaScript, Python. |
| **GNU/Linux Tools** | make, gcc, gdb, kgdb, valgrind. |
| **Tools** | YOCTO, Buildroot, Git, Quilt, Redmine, Vim |
| **Protocols** | Wi-Fi, I2C, SPI, USB, RS-232. Solid understanding of the OSI model and Wi-FI stack (ath11/10/K , mac80211, cfg80211, nl80211) |

## *Projects and Open-Source Contributions*

**Participation in the French Robotics Cup** *May 2021*
*Theme: Sail the World* *Grenoble, France*

**Open Source**

Participation in events and conferences focused on embedded systems and free/open-source software: Paris Embedded, FOSDEM, Hack in Paris, SysDream.

## *Other*

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|---|---|
| **Hobbies** | Motorsports, Sim Racing, Photography, Videos games.<br>Retro Computing, Retrogaming, 3D , Member of the MO5 Association.<br>Mindfulness meditation, Japan culture, travelling |